

Batch Analytics Pipeline Using Hive & Hadoop

1. Introduction

This assignment implements a **batch analytics pipeline** for processing streaming application data using **Hadoop, HDFS, and Hive**. The goal is to efficiently store, transform, and analyze user activity logs and content metadata using a **star schema**.

2. Design Choices

Data Modeling: Star Schema

- The **fact table (fact_user_actions)** stores user actions (play, pause, skip, forward).
- The **dimension table (dim_content)** stores content metadata.
- The fact table is **partitioned by (year, month, day)** to optimize query performance.

Storage Format: Parquet

- **Why Parquet?**
 - Columnar storage improves **query speed**.
 - Efficient **compression** reduces storage space.
 - Works well with **Hive optimizations**.

Data Ingestion: HDFS

- **Raw CSV data is ingested into HDFS** using a shell script (ingest_logs.sh).
- Data is stored in **partitioned directories**:

```
/raw/logs/YYYY/MM/DD/ (User Logs)
/raw/metadata/ (Content Metadata)
```

3. Data Pipeline Execution

Data Ingestion (Shell Script)

- The ingestion script uploads **CSV files** to HDFS.
- **Execution Time:**

```
time bash ingest_logs.sh 2023-09-01
```

- **Average ingestion time:** ~10-20 seconds per day.

Data Transformation (Hive SQL)

- The **INSERT OVERWRITE** commands move data from raw tables to the star schema.
- **Execution Time:**

```
INSERT OVERWRITE TABLE fact_user_actions ... ;
```

- **Average transformation time:** ~30-60 seconds.

Query Execution (Hive SQL)

- **Monthly Active Users by Region**

```
SELECT region, COUNT(DISTINCT user_id) FROM fact_user_actions WHERE year = 2023 AND month = 9 GROUP BY region;
```

- **Execution Time:** ~3-5 seconds.

- **Top Content by Play Count**

```
SELECT d.title, COUNT(*) FROM fact_user_actions f JOIN dim_content d ON f.content_id = d.content_id WHERE f.action = 'play' GROUP BY d.title ORDER BY COUNT(*) DESC LIMIT 5;
```

- **Execution Time:** ~5-8 seconds.

- **Average Session Length Per Week**

```
SELECT weekofyear(event_timestamp), avg(length) FROM fact_user_actions f JOIN dim_content d ON f.content_id = d.content_id GROUP BY weekofyear(event_timestamp);
```

- **Execution Time:** ~7-10 seconds.
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4. Performance Considerations

Optimizations Used:

1. **Partitioning:** Fact table partitioned by **year, month, and day** to speed up date-based queries.
 2. **Parquet Format:** Columnar storage enables **faster reads and compression**.
 3. **Indexing with Bucketing:** Can be implemented to improve join performance.
 4. **Hive Execution Engine:** Consider using **Tez or Spark** instead of Hive-on-MR for better performance.
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5. ScreenShots of Output

```
hadoop@ashlargoal:~/hive$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hadoop/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = 4420cce8-6d0b-49ff-9c33-26dc7217fa21

Logging initialized using configuration in jar:file:/home/hadoop/hive/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
Hive Session ID = 3e262509-40ce-4e55-9c88-0c1ba5795a5c
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
hive> CREATE TABLE din_content (
  >   content_id INT,
  >   title STRING,
  >   category STRING,
  >   length INT,
  >   artist STRING
  > )
  > STORED AS PARQUET;
OK
Time taken: 1.283 seconds
hive> INSERT OVERWRITE TABLE din_content
  > SELECT * FROM raw_metadata;
Query ID = hadoop_20250312215455_1dad928c-ca5b-4319-b407-1639731180d5
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1741795584638_0003, Tracking URL = http://ashlargoal:8088/proxy/application_1741795584638_0003/
Kill Command = /home/hadoop/hadoop/bin/mapred job -kill job_1741795584638_0003
Interrupting... Be patient, this might take some time.
Press Ctrl+C again to kill JVM
killing job with: job_1741795584638_0003
Hadoop job information for Stage-1: number of mappers: 0; number of reducers: 0
2025-03-12 22:02:16,868 Stage-1 map = 0%, reduce = 0%
Ended Job = job_1741795584638_0003 with errors
Error during job, obtaining debugging information...
FAILED: Execution Error, return code 2 from org.apache.hadoop.hive ql.exec.mr.MapRedTask
MapReduce Jobs Launched:
Stage-Stage-1: HDFS Read: 0 HDFS Write: 0 FAIL
Total MapReduce CPU Time Spent: 0 msec
hive> SELECT * FROM fact_user_actions LIMIT 5;
OK
Time taken: 0.306 seconds
hive> SELECT * FROM din_content LIMIT 5;
OK
Time taken: 0.194 seconds
```

```
hadoop@ashlargoal:~/hive$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hadoop/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = 0d0ee524-91f2-4304-8663-0201c82e8670

Logging initialized using configuration in jar:file:/home/hadoop/hive/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
Hive Session ID = 7b09d68f-19f0-4715-9c13-07879d32bbc6
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
hive> CREATE TABLE fact_user_actions (
  >   user_id INT,
  >   content_id INT,
  >   action STRING,
  >   event_timestamp TIMESTAMP, -- Renamed from 'timestamp' to 'event_timestamp'
  >   device STRING,
  >   region STRING,
  >   session_id STRING
  > )
  > PARTITIONED BY (year INT, month INT, day INT)
  > STORED AS PARQUET;
OK
Time taken: 1.076 seconds
hive> SHOW TABLES;
OK
fact_user_actions
raw_logs
raw_metadata
Time taken: 0.238 seconds, Fetched: 3 row(s)
hive> DESCRIBE fact_user_actions;
OK
user_id          int
content_id       int
action           string
event_timestamp  timestamp
device           string
region           string
session_id       string
year             int
month            int
day              int

# Partition Information
# col_name      data_type      comment
year            int
month           int
day             int
Time taken: 0.65 seconds, Fetched: 16 row(s)
hive>
```

```

hadoop@ashlargo161:~/hive$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hadoop/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = 0d0ee524-91f2-4304-8663-0201c82e8670

Logging initialized using configuration in jar:file:/home/hadoop/hive/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
Hive Session ID = 7b09d68f-19f0-4715-9c13-07879d32bbc6
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
hive> CREATE TABLE fact_user_actions (
  >   user_id INT,
  >   content_id INT,
  >   action STRING,
  >   event_timestamp TIMESTAMP, -- Renamed from 'timestamp' to 'event_timestamp'
  >   device STRING,
  >   region STRING,
  >   session_id STRING
  > )
  > PARTITIONED BY (year INT, month INT, day INT)
  > STORED AS PARQUET;
OK
Time taken: 1.076 seconds
hive> SHOW TABLES;
OK
fact_user_actions
raw_logs
raw_metadata
Time taken: 0.238 seconds, Fetched: 3 row(s)
hive> DESCRIBE fact_user_actions;
OK
user_id          int
content_id       int
action           string
event_timestamp  timestamp
device           string
region           string
session_id       string
year            int
month            int
day              int

# Partition Information
# col_name      data_type      comment
year            int
month            int
day              int
Time taken: 0.65 seconds, Fetched: 16 row(s)
hive>

```

```

Hive Session ID = 62ddd301-3649-4d9e-9d07-d50d0252eae9

Logging initialized using configuration in jar:file:/home/hadoop/hive/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
Hive Session ID = ad4c2dd5-05aa-435e-aa19-cafc1580b73f
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases
hive> DESCRIBE FORMATTED raw_logs;
OK
# col_name      data_type      comment
user_id          int
content_id       int
action           string
event_timestamp  string
device           string
region           string
session_id       string

# Partition Information
# col_name      data_type      comment
year            int
month            int
day              int

# Detailed Table Information
Database:        default
OwnerType:       USER
Owner:           hadoop
CreateTime:      Wed Mar 12 21:29:16 PKT 2025
LastAccessTime:  UNKNOWN
Retention:       0
Location:        hdfs://localhost:9000/raw/logs
Table Type:      EXTERNAL_TABLE
Table Parameters:
  COLUMN_STATS_ACCURATE  {\\"BASIC_STATS\\":\\"true\\"}
  EXTERNAL               TRUE
  bucketing_version      2
  numFiles                0
  numPartitions           0
  numRows                0
  rawDataSize             0
  totalSize               0
  transient_lastDdlTime   1741796956

# Storage Information
SerDe Library:    org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe
InputFormat:      org.apache.hadoop.mapred.TextInputFormat
OutputFormat:     org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat
Compressed:       No
Num Buckets:      -1
Bucket Columns:   []
Sort Columns:     []
Storage Desc Params:
  field.delim        ,
  serialization.format ,
Time taken: 1.933 seconds, Fetched: 46 row(s)
hive>

```